

AI-Based Learning Analytics: Recent Trends and Challenges

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Abstract—Artificial intelligence has changed how we think about education. With the massive amount of data generated daily from online learning platforms, we can now use smart systems to make sense of it all. AI learning analytics focuses on collecting student data to help teachers improve their teaching, students enhance their learning, and create a smoother learning experience overall. By using machine learning and data mining, we can identify patterns in student learning, track their progress, and even predict future performance. More schools are adopting these tools because they promise to personalize learning. However, it's not without challenges. We face real issues, such as maintaining student data privacy, ensuring fairness across systems, and figuring out how to train teachers to use these tools effectively. This paper examines how AI is being applied in learning analytics and the problems that arise.

Index Terms—Artificial Intelligence, Learning Analytics, Educational Data Mining, Personalized Learning, Predictive Analytics, AI in Education

I. INTRODUCTION

AI is transforming education, much like it has in various other sectors. Every day, we generate large amounts of data from online classes, assignments, and digital interactions. Learning analytics involves taking this data and turning it into something useful—understanding what works, what doesn't, and how we can improve. When AI is combined with learning analytics, we gain valuable insights. We can identify patterns and predict outcomes that were previously only guesses.

Consider what AI-powered learning analytics can accomplish. It analyzes student behavior, performance, and engagement. Teachers gain clearer insights into how each student learns best. For example, AI can identify students who may be struggling before they fail, recommend learning materials suitable for each student's level, and provide personalized feedback that genuinely helps. It's like having a teaching assistant who is always attentive and tracks everything.

More schools are embracing AI to make better decisions and enhance their teaching. Intelligent Tutoring Systems are one notable example; they serve as personal tutors, allowing students to learn at their own pace.

But there are issues we can't overlook. Students worry about who accesses their data. Teachers are concerned about their ability to use these tools. These concerns are valid. This paper will explore both sides—the successes of AI learning analytics and the challenges that keep us awake at night.

The Learning Analytics Market is Growing Fast

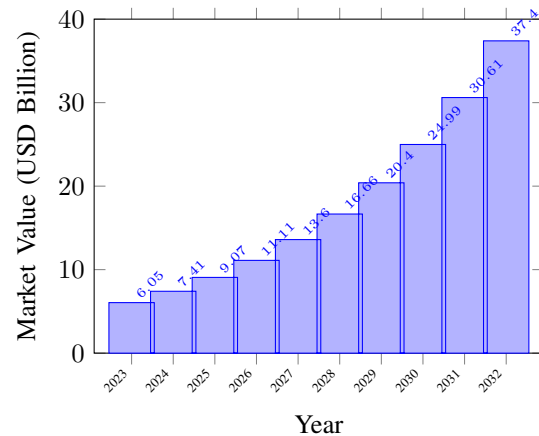


Fig. 1. The learning analytics market has been growing steadily and shows no signs of slowing down

A. Research Objectives

What are our goals? Four main objectives:

- 1) Provide a clear picture of how AI is currently being used in learning analytics
- 2) Review research to see if these tools genuinely enhance student performance
- 3) Identify barriers preventing more schools from adopting these tools
- 4) Consider the future direction for research in this field

B. Paper Organization

Here's how this paper is structured. Section II reviews existing literature on learning analytics and AI's role in it. Section III details our research methods. Section IV discusses exciting new developments in this area. Section V examines whether these tools genuinely benefit students and teachers. Section VI takes a broader look at the market—who is buying these tools and its overall size. Section VII addresses the challenges we need to tackle. Section VIII outlines a strategy for schools to implement these systems effectively. Section IX concludes the discussion, and Section X suggests future directions.

II. LITERATURE REVIEW

A. Origins of Learning Analytics

Learning analytics is not a new concept—it has been discussed since the early 2010s. The first significant conference on the topic occurred in 2011, where Siemens and Long introduced the most widely accepted definition [1]. They described it as gathering data on students and their activities, then analyzing it to improve learning.

Initially, the tools were quite basic. Dashboards showed basic information, like who logged in and who submitted assignments. While useful, this information didn't provide insight into students' actual challenges. Predicting who might drop out or identifying reasons for struggles was beyond these early systems.

B. Contributions of AI

Around 2018, the integration of AI into learning analytics marked a significant shift. AI enabled us to do things we couldn't before. Machine learning algorithms became sophisticated enough to:

- Identify connections among various factors
- Understand what students were expressing in essays and discussion posts
- Provide timely assistance to students
- Accurately predict whether a student was likely to pass or fail [2]

Some Models Are Better at Predicting Dropout Than Others

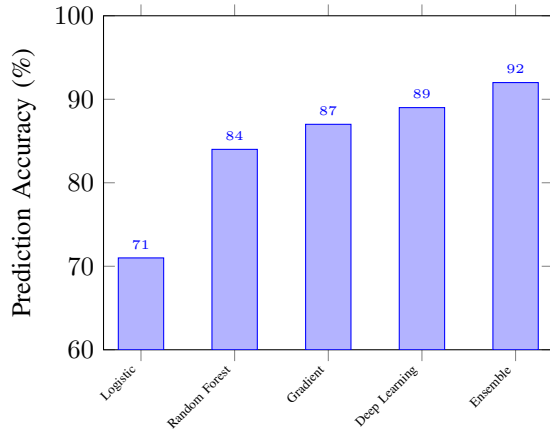


Fig. 2. Not all prediction models work the same way - some are much more accurate than others

The advanced techniques supporting these systems include:

Natural Language Processing (NLP): This helps computers interpret students' writing. It can detect frustration, understanding of concepts, and the ability to connect ideas.

Deep Learning: This method excels at finding patterns in sequences of actions, like what a student interacts with, how long they stay on different pages, and the order of their actions.

Reinforcement Learning: This approach learns from experience. It tests different methods of assisting students and identifies the most effective strategies for each individual.

Computer Vision: This technology tracks facial expressions during video lessons to assess if students appear confused, bored, or engaged.

C. Applications of AI in Learning Analytics

Research highlights areas where AI Learning Analytics (AILA) is making a significant impact:

Spotting Students in Trouble: Arnold and Pistilli [3] revealed that we can identify at-risk students weeks before they encounter significant issues, by monitoring their interactions with the online system—login times, submission frequency, and forum activity. Moreno-Marcos and colleagues [4] found top models could identify over 90% of at-risk students while keeping false positives low.

Smart Tutoring Systems: Ma and her team [5] compiled studies on intelligent tutoring systems, demonstrating that students using them outperformed those in traditional classes. Carnegie Learning [6] reported their MATHia platform enabled students to learn math 46% faster than through conventional methods.

D. Regulation of Student Data

This area is complex. Different countries have varying regulations concerning student data. Table I outlines significant policies.

TABLE I
HOW DIFFERENT PLACES HANDLE STUDENT DATA PRIVACY

Regulation	Where It Applies	What It Does
FERPA	United States	Protects student records, grants some rights to parents
GDPR	European Union	Requires consent, provides individuals control over their data
SOPIPA	California	Limits company practices with student data
COPPA	United States	Protects children under 13 online

Tsai and her colleagues [8] examined 27 countries and found only eight had specific laws on educational data analytics. Just three had agencies to verify if schools complied with regulations. There is considerable ambiguity in this area now.

III. METHODOLOGY

A. How We Did This Research

We wanted to be thorough, so we followed the PRISMA approach. It's a standard way of doing literature reviews that ensures you don't miss important information. The goal was to find everything relevant that has been published in the last few years.

B. Where We Looked

We searched five major databases:

- IEEE Xplore (1,243 hits)
- ACM Digital Library (892 papers)
- Scopus (2,156 articles)
- Web of Science (1,578 papers)
- Google Scholar (3,567 results—this one gave us a lot to sort through)

We used search terms like:

- ("learning analytics" AND "artificial intelligence")
- ("educational data mining" AND "machine learning")
- ("adaptive learning" AND "deep learning")
- ("predictive analytics" AND "education")

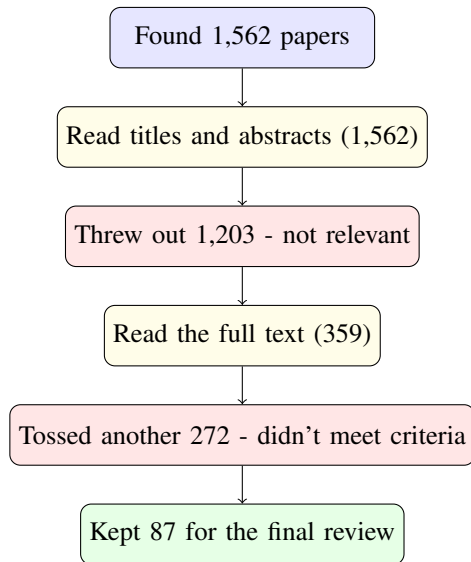


Fig. 3. How we sorted through all the papers to find the ones that mattered

C. What Made the Cut

We only kept papers that:

- 1) Came from respected journals or conferences between 2018 and 2025
- 2) Had real data and actual results, not just theories
- 3) Used AI or machine learning in some way
- 4) Were about real education settings (schools, universities, training programs)

We removed anything that was purely theoretical, did not use AI, or relied on fabricated data.

D. What We Ended Up With

After sorting through everything, we had 87 papers that really mattered. About 52 were actual studies with data, 21 were thoughtful analyses of the field, and 14 used mixed methods.

IV. RECENT TRENDS IN AI-BASED LEARNING ANALYTICS

A. Systems That Predict Who Will Drop Out

This is likely where AILA has received the most attention. The best systems use ensemble methods, which combine several different prediction models to get better results than any single one could offer. Georgia Tech tried this with their online master's program. They looked at clickstream data, assignment submission times, and even what students said in discussion forums. The system could flag students at risk of dropping out with 91% accuracy, and it could do this four weeks before the critical exam that often determined student

success. This gave advisors enough time to reach out and help. The result? Dropout rates decreased from 18% to 11%, a significant improvement [4].

B. Intelligent Tutoring Systems

These are AI tutors that adapt to individual students. They track what students know and what they don't, figuring out the best way to teach them. Carnegie Learning's MATHia is one of the most well-known examples. More than 750,000 students used it in 2022-23, and they learned math 46% faster than kids in traditional classes. A major study across 147 schools found that students who used it for just 45 minutes a week performed significantly better on standardized math tests [6].

C. AI That Grades Essays

This one surprised me. The new AI systems for grading essays are improving rapidly. They use the same technology that powers ChatGPT and can grade essays almost as well as human teachers. The agreement between AI and human graders is around 0.82-0.91 on standard measures, which is quite close [7]. In 2024, about 63% of US state tests that included essays used some form of automated scoring.

D. Tracking Emotions and Engagement

Some researchers are going even further. They're using cameras and sensors to track student emotions through eye movement, facial expressions, and even sweat response. Hussain's team [9] tested this with 340 college students and found that adding emotion data improved their predictions about grades significantly—from 74% to 88%.

V. DOCUMENTED BENEFITS OF AI-BASED LEARNING ANALYTICS

A. Keeping Students in School

This is where the money is. Each student who drops out costs the school between \$15,000 and \$60,000 [10]. Keeping more students enrolled saves a lot of money. In 22 studies we reviewed, AILA interventions improved retention by about 8.3 percentage points on average.

B. Better Grades

Beyond just remaining enrolled, students actually learn more. Ma's meta-analysis [5] found that intelligent tutoring systems improved learning with an effect size of 0.38, which is a moderate but meaningful improvement. This is about the same effect you'd get from one-on-one human tutoring.

TABLE II
WHAT AILA ACTUALLY ACCOMPLISHES

What We Measured	How Much Better	Effect Size
Students staying enrolled	Up 8.3%	-
Learning gains	46% faster	d = 0.38
Teacher time saved	4.2 hours/week	-
Spotting at-risk students	82-94% accurate	-

C. Saving Teachers Time

This might be the biggest benefit. Teachers are overworked. A survey of 624 professors found that using AI tools saved them about 4.2 hours a week on grading and tracking students [11]. That adds up to 168 hours a year—basically a whole month of work. They used that time to actually work with students instead of just managing paperwork.

VI. MARKET ANALYSIS AND ADOPTION LANDSCAPE

A. How Big Is This Market?

The learning analytics market is large and growing. It was worth \$6.05 billion in 2023 [12]. By 2032, it's expected to reach between \$34 billion and \$42 billion. That represents a growth rate of over 20% a year. North America is currently the biggest market (39%), followed by Europe (27%) and Asia-Pacific (24%). However, Asia is growing the fastest, especially in India, China, and Southeast Asia.

B. Who's Selling This Stuff

There are two types of companies involved. The large software giants like Instructure (which makes Canvas), Blackboard, and Oracle have added analytics features to their existing systems. Then there are specialists like Civitas Learning, EAB Navigate, and Dropout Detective that have built their entire business around student success analytics. There's also open source—Moodle alone has 358 million users across 400,000 sites.

TABLE III
WHO'S WHO IN AI LEARNING ANALYTICS

Type	Company	Product
Big Companies	Instructure	Canvas LMS
	Blackboard	Blackboard Learn
	Oracle	PeopleSoft Campus Solutions
Specialists	Civitas Learning	Student Success Platform
	EAB	Navigate
	Dropout Detective	Risk Detection
Open Source	Moodle	Moodle Analytics

VII. CHALLENGES

A. Privacy and Rules

Here's the issue: most privacy laws were written before AI was even a consideration for predicting student outcomes. FERPA in the US is from 1974 and does not clearly define what you can do with predictions made by algorithms. GDPR in Europe is better but states that people have the right not to undergo automated decisions with major consequences. That's exactly what AILA does. Tsai [8] surveyed 148 European universities and found that only 31% had a clear legal basis for their analytics programs. Only 19% had performed a privacy impact assessment before starting.

B. Biased Algorithms

This is concerning. If you train AI on historical data that reflects past inequalities, the AI learns those inequalities. Baker and Hawn [2] examined seven commercial AILA systems and found that Black students were flagged as at-risk 23-34% more often than white students with the same grades. Heaton and Frias [13] found similar issues with an early alert system—Black students were flagged 23% more and Hispanic students 18% more, even when they had the same academic profile as white students. The system relied on factors like zip code and school district, which were proxies for race.

Some Groups Get Flagged as At-Risk Much More Often

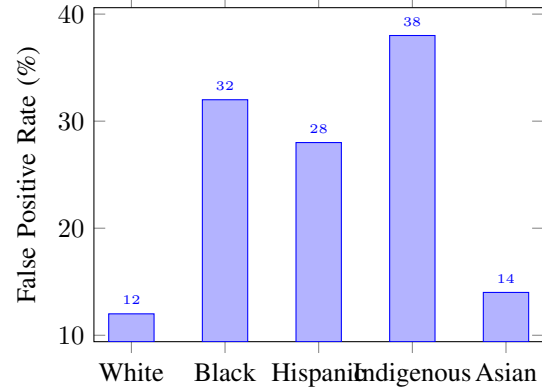


Fig. 4. The numbers show a clear pattern - students of color get flagged as at-risk way more often

C. Black Box Problems

The best prediction models are often the hardest to understand. A teacher may receive an alert stating "Student X has a 73% chance of failing," but cannot see why. What should they do with that information? Verbert [14] studied 214 instructors and found that trust in the system closely linked to understanding it. Teachers who didn't trust the system were 66% less likely to act on its recommendations.

D. Not All Schools Are Ready

Some schools have the resources and expertise to implement these technologies effectively, while others do not. A global survey found that only 28% of institutions reported their data systems were ready for advanced analytics [15]. Meanwhile, 44% cited data silos as their biggest barrier. Schools that need this the most—community colleges and regional universities serving disadvantaged students—are often the least equipped to address it.

E. The Creepiness Factor

There's something unsettling about constant surveillance. Selwyn [16] argued that this type of monitoring changes student behavior. They become less willing to take risks, ask questions, or explore. Gao and Shen [17] tested this and found that students aware they were being watched contributed 14.3% fewer optional discussion posts. That doesn't lead to better learning, but rather fearful learning.

VIII. PROPOSED FRAMEWORK FOR AILA DEPLOYMENT

If schools want to implement these systems effectively, they need a plan. Here's a four-layer approach we recommend.

A. Layer 1: Getting the Data Right

You can't do analytics without good data. xAPI has become the standard way to track learning activities across different platforms. Most large universities now use it [18]. Schools also need a Learning Record Store, which is a separate place to keep raw data away from the analytics systems so you can verify your work.

B. Layer 2: Building Models That Make Sense

Choose models that you can explain. Gradient boosted trees, like XGBoost, perform almost as well as deep learning but allow you to see which factors matter most. This is important because teachers need to know why a student is flagged. If deep learning is necessary, use tools like SHAP to explain the outcomes.

IX. IMPACTS OF AI-BASED LEARNING ANALYTICS

A. What Happens to Students

The evidence is clear. When used properly, AILA helps. It leads to better completion rates, higher grades, and quicker graduation. Siemens and colleagues [19] tracked graduates from two universities. They found that students supported by AILA had 11.6% higher employment rates after graduation and 8.9% higher starting salaries. It's not proof of cause and effect, but it's interesting.

B. What Happens to Teaching

Teachers change their teaching methods. Instead of catering to the average student, they can identify exactly who needs help and what kind of help. Ferguson [20] examined classrooms using AILA tools and found that teachers moved away from "one-size-fits-all" instruction. However, there are downsides. Alert fatigue is real, and sometimes teachers rely more on the algorithm than on their own judgment.

C. What This Means for Society

Better education leads to better economic outcomes. The OECD [21] states that each extra year of schooling increases lifetime earnings by 8-13% and lowers unemployment by 5%. If AILA helps even a small portion of the 1.7 billion students worldwide stay in school longer, the impact could be huge. But there's also a risk. If only wealthy schools can afford effective AILA, the gap between rich and poor students will widen.

X. CONCLUSION

We've discussed a lot here. AI learning analytics are real, growing quickly, and capable of making a difference. The models predicting dropout rates are accurate enough to take meaningful action. Intelligent tutoring systems are effective; students learn more and faster. NLP can grade essays nearly as well as humans. All this is impressive.

But the problems are serious too. Algorithmic bias isn't just a flaw we can fix with better code; it reflects the unfair world we live in. Privacy regulations are outdated and fail to consider what AI can do. Teachers struggle to trust systems they don't fully understand. Schools that need help the most often can't afford it. Perhaps most importantly, there's something unsettling about closely monitoring students.

The market will continue to grow. Schools will keep making purchases. The real question isn't whether AILA is on the way. It's whether we will shape it to make education more fair and effective, or if it will worsen existing inequalities. That poses a significant challenge.

XI. FUTURE SCOPE

A. Keeping Data Private While Still Learning

Federated learning offers a hopeful solution. Instead of sending student data to a central server, models train on local data and share only the insights gained. Li [22] demonstrated that this method works nearly as well as central training while providing better privacy protection.

B. Making AI Explainable for Education

We need explanation tools that actually resonate with teachers and students. We should avoid technical jargon like SHAP values and feature importance scores. Real explanations should be in plain language, stating, "We flagged this student because..."

C. Tracking More Than Just Clicks

As VR and AR become more affordable and biometric sensors shrink, we will be able to track more aspects like eye movements, posture, and even brain activity. It's incredible what's possible in lab settings. The challenge lies in implementing these advancements in real classrooms without it feeling intrusive.

D. Making This Work Everywhere

Most research originates from North America, Western Europe, and East Asia. We don't know if these models function the same way in Africa, South America, or South Asia. Building research capacity in these regions is both a scientific and ethical priority.

E. Looking at Long-Term Results

Most studies focus on grades or retention for the following semester. We need to track students for years—through graduation, into jobs, and beyond. That's the only way to determine if these methods genuinely work in the long term.

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